Totally ordered abelian groups

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1 Ordered sets

Definition 1.1. Suppose S is a set. An *order* on S is a relation $\geqslant$ on S satisfying the following conditions.

(1) (Reflexive) For any $x \in S$ we have $x \geqslant x$.

(2) (Transitive) If $x \geqslant y$ and $y \geqslant z$, then $x \geqslant z$.

(3) (Antisymmetric) If $x \geqslant y$ and $y \geqslant x$, then $x = y$.

An order is *total* if, for any x and y in S, either $x \geqslant y$ or $y \geqslant x$.

Definition 1.2. (a) A *ordered set* is a set equipped with an order.

(b) A *totally ordered set* is a set equipped with a total order.

Definition 1.3. Suppose S and T are ordered sets and $\phi : S \to T$ is a function.

(a) ϕ is *order preserving* if $x \geqslant y$ in S implies $\phi(x) \geqslant \phi(y)$ in T.

(b) ϕ is *order reversing* if $x \geqslant y$ in S implies $\phi(x) \leqslant \phi(y)$ in T.

(c) ϕ is an *order embedding* if $x \geqslant y$ in S if and only if $\phi(x) \geqslant \phi(y)$ in T.

(d) ϕ is an *order isomorphism* if it is both a bijection and an order embedding.

Lemma 1.4. *Suppose S and T are ordered sets and $\phi : S \to T$ is an order embedding. Then ϕ is injective.*

Proof. Suppose $\phi(x) = \phi(y)$. Then we have $\phi(x) \geqslant \phi(y)$ and $\phi(y) \geqslant \phi(x)$, so ϕ being an order embedding implies that $x \geqslant y$ and $y \geqslant x$. Antisymmetry in S then forces $x = y$. □

Lemma 1.5. *Suppose S and T are ordered sets and $\phi : S \to T$ is a function. If S is totally ordered, then the following are equivalent.*

(a) ϕ is an order embedding.

(b) ϕ is injective and order preserving.

Proof. It follows from lemma 1.4 and definitions that (a) implies (b). Conversely, suppose ϕ is injective and order preserving. Suppose we have $x, y \in S$ such that $\phi(x) \geqslant \phi(y)$. We want to show that $x \geqslant y$. Since S is totally ordered, we can assume for a contradiction that $x \lneqq y$. Since ϕ is order preserving, this means that $\phi(x) \leqslant \phi(y)$. Antisymmetry in T forces $\phi(x) = \phi(y)$, and ϕ being injective then implies $x = y$, contradicting our assumption that $x \lneqq y$. □

Definition 1.6. Suppose S is an ordered set. For $a, b \in S$, we define the *interval* $[a, b]$ by

$$[a, b] = \{x \in S : a \leqslant x \leqslant b \text{ or } b \leqslant x \leqslant a\}.$$

Definition 1.7. Let S be an ordered set. A subset $T \subseteq S$ is *convex* if, whenever $a, b \in T$, then $[a, b] \subseteq T$.

Lemma 1.8. *Let S_1 and S_2 be ordered sets. Then there exists on the disjoint union $S_1 \sqcup S_2$ a unique order satisfying both of the following conditions.*

(1) The inclusions of S_1 and S_2 are both order embeddings.

(2) *All elements of* S_1 *are less than all elements of* S_2.

Proof. Omitted. □

Definition 1.9. If S_1 and S_2 are ordered sets, their *ordered sum* $S_1 + S_2$ is the disjoint union $S \sqcup S_2$ equipped with the unique order from lemma 1.8.

Lemma 1.10. *If* S_1 *and* S_2 *are totally ordered sets, then so is* $S_1 + S_2$.

Proof. Omitted. □

2 Totally ordered abelian groups

Definition 2.1. Suppose Γ is an abelian group. A relation $R \subseteq \Gamma \times \Gamma$ is *translation invariant* if, whenever $(a, b) \in R$, then

$$(a + x, b + x) \in R$$

for all $x \in \Gamma$.

Lemma 2.2. *Let* Γ *be an abelian group. There is an injective map*

$$\{\textit{translation invariant total orders on } \Gamma\} \longrightarrow \{\textit{submonoids of } \Gamma\}$$

which carries a translation invariant total order $\geqslant$ *on* Γ *to the submonoid* $\Gamma_{\geqslant 0} = \{x \in \Gamma : x \geqslant 0\}$. *Its image consists of the submonoids* $S \subseteq \Gamma$ *satisfying both of the following conditions.*

(1) *If* $x \in S$ *and* $-x \in S$, *then* $x = 0$.

(2) *For every* $x \in \Gamma$, *either* $x \in S$ *or* $-x \in S$.

Proof. Suppose we have a translation invariant total order $\geqslant$ on Γ and $a, b \in \Gamma_{\geqslant 0}$. Since $a \geqslant 0$, translation-invariance guarantees that

$$a + b \geqslant b \geqslant 0.$$

Thus $\Gamma_{\geqslant 0}$ is a submonoid of Γ and the map $\geqslant \mapsto \Gamma_{\geqslant 0}$ is well-defined.

Suppose $\geqslant$ and $\geqslant'$ are both translation invariant total orders on Γ such that $\Gamma_{\geqslant 0} = \Gamma_{\geqslant' 0}$. Suppose we have $a, b \in \Gamma$ such that $a \geqslant b$. By translation invariance of $\geqslant$, we have $a - b \geqslant 0$, so $a - b \in \Gamma_{\geqslant 0} = \Gamma_{\geqslant' 0}$. In other words, $a - b \geqslant' 0$, so by translation invariance of $\geqslant'$, have $a \geqslant' b$. Thus $a \geqslant b$ implies $a \geqslant' b$, and the converse implication is entirely symmetric. This shows that the map $\geqslant \mapsto \Gamma_{\geqslant 0}$ is injective.

Suppose we are given a submonoid $S \subseteq \Gamma$. We define a relation $\geqslant$ on Γ by declaring $x \geqslant y$ if and only if $x - y \in S$. The fact that S is a submonoid implies automatically that $\geqslant$ is reflexive, transitive, and translation invariant. It follows that $\geqslant$ is antisymmetric if and only if S satisfies condition (1), and that $\geqslant$ is total if and only if S satisfies condition (2) holds. □

Lemma 2.3. *Let Γ be an abelian group and suppose $\geqslant$ and $\geqslant'$ are two total orders on Γ. If $\Gamma_{\geqslant 0} \subseteq \Gamma_{\geqslant' 0}$, then $\geqslant$ and $\geqslant'$ are equal.*

Proof. By the injecitivity assertion of lemma 2.2, it is sufficient to show that $\Gamma_{\geqslant 0} = \Gamma_{\geqslant' 0}$. Suppose for a contradiction that $x \in \Gamma_{\geqslant' 0} \setminus \Gamma_{\geqslant 0}$. Then $-x \in \Gamma_{\geqslant 0} \subseteq \Gamma_{\geqslant' 0}$, so x and $-x$ are both contained in $\Gamma_{\geqslant' 0}$. This means that $x = 0$, which contradicts our assumption that $x \notin \Gamma_{\geqslant 0}$. □

Definition 2.4. A *totally ordered abelian group* is an abelian group Γ equipped with a translation invariant total order.

Lemma 2.5. *Suppose Γ is a totally ordered abelian group. Then Γ is torsion free.*

Proof. Suppose $x \in \Gamma$ and $x \gneqq 0$. By translation invariance, we see that

$$\cdots \gneqq 3x \gneqq 2x \gneqq x \gneqq 0.$$

Indeed, if there were some positive integer n such that $(n+1)x = nx$, then by subtracting nx from both sides we would find that $x = 0$. In particular, we see that x cannot be torsion. Similarly, if $x \lneqq 0$, then

$$0 \gneqq x \gneqq 2x \gneqq 3x \gneqq \cdots,$$

and again x cannot be torsion. □

Lemma 2.6. *Suppose Γ and Γ' are totally ordered abelian groups and $\phi : \Gamma \to \Gamma'$ is a homomorphism of abelian groups. Then the following are equivalent.*

(a) *ϕ is order preserving.*

(b) $\phi(\Gamma_{\geqslant 0}) \subseteq \Gamma'_{\geqslant 0}$.

Proof. If $x \geqslant y$ in Γ, then $x - y \in \Gamma_{\geqslant 0}$, so $\phi(x-y) = \phi(x) - \phi(y) \in \phi(\Gamma_{\geqslant 0})$. The statement follows immediately. □

Lemma 2.7. *Suppose Γ and Γ' are totally ordered abelian groups and $\phi : \Gamma \to \Gamma'$ is a homomorphism of abelian groups. Then the following are equivalent.*

(a) *ϕ is an order embedding.*

(b) $\Gamma_{\geqslant 0} = \phi^{-1}(\Gamma'_{\geqslant 0})$.

Proof. Suppose ϕ is an order embedding. We know from lemma 2.6 that $\Gamma_{\geqslant 0} \subseteq \phi^{-1}(\Gamma'_{\geqslant 0})$. For the reverse inclusion, suppose $x \in \phi^{-1}(\Gamma'_{\geqslant 0})$. In other words, we have $\phi(x) \geqslant 0$. We would like to show that $x \geqslant 0$, so suppose for a contradiction that $x \lneqq 0$. Since ϕ is order preserving, this implies that $\phi(x) \leqslant \phi(0) = 0$, so antisymmetry forces $\phi(x) = 0$. But ϕ is injective by lemma 1.4, so $x = 0$, which contradicts our assumption that $x \lneqq 0$.

Conversely, suppose $\Gamma_{\geqslant 0} = \phi^{-1}(\Gamma'_{\geqslant 0})$. It follows from lemma 2.6 that ϕ is order preserving. Suppose $x, y \in \Gamma$ are such that $\phi(x) \geqslant \phi(y)$. Then $\phi(x) - \phi(y) = \phi(x-y) \in \Gamma'_{\geqslant 0}$, or, in other words, $x - y \in \phi^{-1}(\Gamma'_{\geqslant 0})$. Since $\phi^{-1}(\Gamma'_{\geqslant 0}) = \Gamma_{\geqslant 0}$, we conclude that $x \geqslant y$. □

3 Constructions

Example 3.1. The real numbers $\mathbb{R}$, regarded as an abelian group under addition and endowed with the standard order, form a totally ordered abelian group.

Example 3.2. Regard $\mathbb{R}[\epsilon] = \{a_0 + a_1\epsilon + \cdots + a_n\epsilon^n : a_0, \ldots, a_n \in \mathbb{R}\}$ as an abelian group under addition. Then the submonoid

$$S = \{a_k\epsilon^k + a_{k+1}\epsilon + \cdots + a_n\epsilon^n : a_k \gneq 0\}$$

defines a translation invariant total order on $\mathbb{R}[\epsilon]$ under the correspondence of lemma 2.2. In the resulting totally ordered abelian group $\mathbb{R}[\epsilon]$, the element ϵ "behaves like a positive infinitesimal." For example, we have

$$0 \lneq \cdots \lneq \epsilon^2 \lneq \epsilon \lneq a$$

for any $a \in \mathbb{R}$.

Example 3.3. If Γ is a totally ordered abelian group and $\Gamma' \subseteq \Gamma$ is any subgroup, then Γ' is a totally ordered abelian group with the induced total order.

Example 3.4. Suppose Γ is a totally ordered abelian group. It is torsion free by lemma 2.5, and there exists on the divisible closure $\mathbb{Q} \otimes \Gamma$ a unique translation invariant total order such that $\Gamma \hookrightarrow \mathbb{Q} \otimes \Gamma$ is an order embedding. It satisfies

$$(\mathbb{Q} \otimes \Gamma)_{\geqslant 0} = \{x \in \mathbb{Q} \otimes \Gamma \ : \ nx \in \Gamma_{\geqslant 0} \text{ for some } n \geqslant 1\}.$$

Proof of existence. We use lemma 2.2. Observe that the set

$$S = \{x \in \mathbb{Q} \otimes \Gamma \ : \ nx \in \Gamma_{\geqslant 0} \text{ for some } n \geqslant 1\}$$

is a submonoid of $\mathbb{Q} \otimes \Gamma$. Indeed, it clearly contains zero, and if we have $x, y \in S$, then there exist $m, n \geqslant 1$ such that $mx, ny \in \Gamma_{\geqslant 0}$. Then

$$mn(x + y) = n(mx) + m(ny) \in \Gamma_{\geqslant 0}$$

showing that $x + y \in S$. If $x \in S$ and $-x \in S$, then there exist $m, n \geqslant 1$ such that $mx \in \Gamma_{\geqslant 0}$ and $-nx \in \Gamma_{\geqslant 0}$. Then mnx and $-mnx$ are both in $\Gamma_{\geqslant 0}$, which means that $mnx = 0$. Since $\mathbb{Q} \otimes \Gamma$ is clearly torsion free, this means that $x = 0$, proving that condition (1) is satisfied. Finally, to check that condition (2) is satisfied, suppose that we have an arbitrary element $x \in \mathbb{Q} \otimes \Gamma$. Then there exists some integer $n \geqslant 1$ such that $nx \in \Gamma$. Then either $nx \in \Gamma_{\geqslant 0}$ or $-nx \in \Gamma_{\geqslant 0}$. In the first case, we have $x \in S$, and in the second, we have $-x \in S$. Thus there exists a unique translation invariant total order $\geqslant$ on $\mathbb{Q} \otimes \Gamma$ such that $(\mathbb{Q} \otimes \Gamma)_{\geqslant 0} = S$. $\square$

Proof of uniqueness. Let S be the submonoid described in the proof of existence above, and suppose $\geqslant$ is an arbitrary translation invariant total order of $\mathbb{Q} \otimes \Gamma$ such that $\Gamma \hookrightarrow \mathbb{Q} \otimes \Gamma$ is an order embedding. Then $(\mathbb{Q} \otimes \Gamma)_{\geqslant 0} \cap \Gamma = \Gamma_{\geqslant 0}$ by lemma 2.7. We claim that this implies that $S = (\mathbb{Q} \otimes \Gamma)_{\geqslant 0}$, and then the injectivity assertion of lemma 2.2 completes the proof of uniqueness.

To see that $S = (\mathbb{Q} \otimes \Gamma)_{\geqslant 0}$, suppose first that $x \in S$. Then $nx \in \Gamma_{\geqslant 0} = (\mathbb{Q} \otimes \Gamma)_{\geqslant 0} \cap \Gamma$ for some $n \geqslant 1$. In particular, $nx \in (\mathbb{Q} \otimes \Gamma)_{\geqslant 0}$, so clearly we must have $x \in (\mathbb{Q} \otimes \Gamma)_{\geqslant 0}$. On the other hand, if $x \in (\mathbb{Q} \otimes \Gamma)_{\geqslant 0}$, there exists some integer $n \geqslant 1$ such that $nx \in \Gamma$. Then $nx \in (\mathbb{Q} \otimes \Gamma)_{\geqslant 0} \cap \Gamma = \Gamma_{\geqslant 0}$, so $x \in S$ by definition of S. □

Example 3.5. Suppose I is a totally ordered index set and $(\Gamma_i)_{i \in I}$ a family of totally ordered abelian groups. For any $\gamma = (\gamma_i)_{i \in I} \in \prod_{i \in I} \Gamma_i$, we define

$$\operatorname{supp}(\gamma) = \{i \in I : \gamma_i \neq 0\}.$$

The *Hahn product* of the family $(\Gamma_i)_{i \in I}$ is defined by

$$\prod_{i \in I}' \Gamma_i = \left\{ \gamma \in \prod_{i \in I} \Gamma_i \ : \ \operatorname{supp}(\gamma) \text{ is well ordered} \right\}.$$

This is a subgroup of $\prod_{i \in I} \Gamma_i$, and is endowed with a total order $\geqslant$ corresponding (under lemma 2.2) to the submonoid

$$\left\{ \gamma = (\gamma_i)_{i \in I} \in \prod_{i \in I}' \Gamma_i \ : \ \gamma = 0 \text{ or } \gamma_j \geqslant 0 \text{ for } j = \min \operatorname{supp}(\gamma) \right\}. \tag{3.6}$$

This total order $\geqslant$ is called the *lexicographic order* on the Hahn product.

Proof that the Hahn product is a subgroup. It is clear that the Hahn product contains the zero element. Suppose we have $\gamma = (\gamma_i)_{i \in I}$ and $\gamma' = (\gamma'_i)_{i \in I}$ such that $\operatorname{supp}(\gamma)$ and $\operatorname{supp}(\gamma')$ are well ordered. Let S be a nonempty subset of $\operatorname{supp}(\gamma + \gamma')$, and suppose for a contradiction that it has no minimal element. Then we can find an infinite, strictly decreasing sequence $i_0 \gneqq i_1 \gneqq \cdots$ in S. In other words, we have $\gamma_{i_k} + \gamma'_{i_k} \neq 0$ for all $k = 0, 1, \ldots$.

Since $\operatorname{supp}(\gamma)$ is well-ordered, no infinite subsequence of $i_0 \gneqq i_1 \gneqq \cdots$ can be contained in $\operatorname{supp}(\gamma)$, which means that there exists some K sufficiently large such that $\gamma_{i_k} = 0$ for all $k \geqslant K$. For the same reasons, we can make K larger if necessary in order to have that $\gamma'_{i_k} = 0$ for all $k \geqslant K$ as well. But this clearly forces $\gamma_{i_k} + \gamma'_{i_k} = 0$, which is a contradiction. □

Proof that the lexicographic order is a total order on the Hahn product. We need to verify that the submonoid in equation (3.6) satisfies the conditions of lemma 2.2. Condition (2) is clear. For condition (1), suppose both $\gamma = (\gamma_i)_{i \in I}$ and its negative are in the submonoid of equation (3.6). Suppose there exists $j \in \operatorname{supp}(\gamma)$. Since $\operatorname{supp}(\gamma) = \operatorname{supp}(-\gamma)$, we see that $\gamma_j \geqslant 0$ and $-\gamma_j \geqslant 0$, so $\gamma_j = 0$. This contradicts our choice of j, so in fact $\operatorname{supp}(\gamma) = \emptyset$. In other words, $\gamma = 0$. □

4 Convex subgroups

Definition 4.1. Suppose Γ is a totally ordered abelian group. A *convex subgroup* (or an *isolated subgroup*) of Γ is a subgroup which is convex in the sense of definition 1.7.

Lemma 4.2. *Suppose Γ is a totally ordered abelian group and Δ is a subgroup. Then the following are equivalent.*

(a) Δ is convex.

(b) Whenever $a \geqslant x \geqslant 0$ and $a \in \Delta$, then $x \in \Delta$ also.

Proof. Suppose we have $a \geqslant x \geqslant b$ where $a, b \in \Delta$. By translation invariance, we have

$$a - b \geqslant x - b \geqslant 0$$

and $a - b \in \Delta$, so the condition in (b) implies that $x - b \in \Delta$. But since $b \in \Delta$, this means that $x \in \Delta$ as well. □

Lemma 4.3. *Suppose $\phi : \Gamma \to \Gamma'$ is an order preserving homomorophism of totally ordered abelian groups. Then $\ker(\phi)$ is convex.*

Proof. We use the criterion of lemma 4.2. Suppose we have $a \geqslant x \geqslant 0$ in Γ with $a \in \ker(\phi)$. Then

$$0 = \phi(a) \geqslant \phi(x) \geqslant 0,$$

so antisymmetry forces $\phi(x) = 0$. □

Lemma 4.4. *Suppose Γ is a totally ordered abelian group, Δ is a convex subgroup of Γ, and Δ' is a convex subgrup of Δ. Then Δ' is a convex subgroup of Γ.*

Proof. We use the criterion of lemma 4.2. Suppose we have $a \geqslant x \geqslant 0$ in Γ where $a \in \Delta'$. Since $\Delta' \subseteq \Delta$ and Δ is a convex subgroup of Δ, we see that $x \in \Delta$. Then since Δ' is convex in Δ, we have $x \in \Delta'$ also. □

Lemma 4.5. *Let Γ be a totally ordered abelian group and suppose Δ and Δ' are two convex subgroups of Γ. Then either $\Delta \subseteq \Delta'$ or $\Delta' \subseteq \Delta$.*

Proof. Suppose $\Delta \not\subseteq \Delta'$. Then there exists $a \in \Delta \setminus \Delta'$, and by replacing a with $-a$ if necessary, we can assume $a \geqslant 0$. Suppose $x \in \Delta'$. We claim that $x \in \Delta$. By replacing x with $-x$ if necessary, we can assume that $x \geqslant 0$. If $x \geqslant a$, then we would have $x \geqslant a \geqslant 0$ and convexity of Δ' would force $a \in \Delta'$, contradicting our choice of a. Thus we must have $a \geqslant x \geqslant 0$, and convexity of Δ now implies that $x \in \Delta$. □

Lemma 4.6. *Suppose Γ is a totally ordered abelian group and $(\Delta_i)_{i \in I}$ is a family of convex subgroups of Γ. Then $\Delta = \bigcap_{i \in I} \Delta_i$ is a convex subgroup of Γ.*

Proof. We use the criterion of lemma 4.2. Suppose we have $a \geqslant x \geqslant 0$ in Γ where $a \in \Delta$. Then $a \in \Delta_i$ for all i, so $x \in \Delta_i$ for all i, so $x \in \Delta$. □

Lemma 4.7. *Suppose Γ is a totally ordered abelian group and $(\Delta_i)_{i \in I}$ is a nonempty family of convex subgroups of Γ. Then $\Delta = \bigcup_{i \in I} \Delta_i$ is a convex subgroup of Γ.*

Proof. Clearly $0 \in \Delta$. If $a, b \in \Delta$, then there exist $i, j \in I$ such that $a \in \Delta_i$ and $b \in \Delta_j$. By lemma 4.5), we can assume without loss of generality that $\Delta_i \subseteq \Delta_j$. Then $a - b \in \Delta_j \subseteq \Delta$ also, proving that Δ is a subgroup. To see that it is convex, we use the criterion of lemma 4.2. Suppose we have $a \geqslant x \geqslant 0$ in Γ where $a \in \Delta$. Then there exists i such that $a \in \Delta_i$, so then $x \in \Delta_i \subseteq \Delta$ also. □

Lemma 4.8. *Let Γ be a totally ordered abelian group and Δ a convex subgroup. If Γ is divisible, then so is Δ.*

Proof. Suppose that $x \in \Gamma$ is some element such that $nx \in \Delta$ for some $n \geqslant 1$. We want to show that $x \in \Delta$. We can assume without loss of generality that $x \geqslant 0$. Translation invariance implies that

$$\cdots \geqslant nx \geqslant (n-1)x \geqslant \cdots \geqslant 2x \geqslant x \geqslant 0,$$

so $x \in [0, nx]$. Convexity of Δ implies that $x \in \Delta$. □

Definition 4.9. Let Γ be a totally ordered abelian group. Given any subset $S \subseteq \Gamma$, we define the *convex subgroup of Γ generated by* S to be the intersection of all convex subgroups of Γ containing S. By lemma 4.6, this is the smallest convex subgroup of Γ containing S.

Lemma 4.10. *Suppose Γ is a totally ordered abelian group, H is a subgroup of Γ and Δ is the convex subgroup of Γ generated by H. Then*

$$\Delta = \bigcup_{h \in H_{\geqslant 0}} [-h, h].$$

Proof. It is clear that $S = \bigcup_{h \in H_{\geqslant 0}} [-h, h]$ must be contained in every convex subgroup containing H, so $S \subseteq \Delta$. To prove that $S = \Delta$, we need to show that S is actually a convex subgroup. It clearly contains zero and is stable under negation. If we have $x, x' \in S$, then there exist $h, h' \in H_{\geqslant 0}$ such that $x \in [-h, h]$ and $x' \in [-h', h']$. Then

$$h + h' \geqslant x + x' \geqslant -h - h'.$$

In other words, we have $x + x' \in [-(h + h'), h + h'] \subseteq S$, showing that S is stable under addition as well. Finally, to see that S is convex, we use the criterion of lemma 4.2. Suppose we have $a \geqslant x \geqslant 0$ with $a \in S$. Then there exists some $h \in H_{\geqslant 0}$ such that $a \in [-h, h]$. But then clearly $x \in [-h, h]$ also, so $x \in S$ as well. □

5 Ordered set of convex subgroups

Definition 5.1. Let Γ be a totally ordered abelian group. We write $\mathrm{Cnv}(\Gamma)$ for the set of all convex subgroups of Γ. By lemma 4.5, this set is totally ordered by inclusion; evidently 0 and Γ are the minimum and maximum elements of $\mathrm{Cnv}(\Gamma)$, respectively. We write $\mathrm{Cnv}^\circ(\Gamma)$ for $\mathrm{Cnv}(\Gamma) \setminus \{\Gamma\}$.

Lemma 5.2. *Suppose $\phi : \Gamma \to \Gamma'$ is an order preserving homomorphism between totally ordered abelian groups. Then $\Delta' \mapsto \phi^{-1}(\Delta')$ is an order preserving map $\phi^{-1} : \mathrm{Cnv}(\Gamma') \to \mathrm{Cnv}(\Gamma)$.*

Proof. Suppose Δ' is a convex subgroup of Γ' and suppose $a, b \in \phi^{-1}(\Delta')$ and that $a \leqslant x \leqslant b$ for some $x \in \Gamma$. Since ϕ is order preserving, we have $\phi(a) \leqslant \phi(x) \leqslant \phi(b)$ in Γ'. Since $\phi(a), \phi(b) \in \Delta'$ and Δ' is convex, this means that $\phi(x) \in \Delta'$, so $x \in \phi^{-1}(\Delta')$. In other words, $\phi^{-1}(\Delta')$ is a convex subgroup of Γ. The fact that $\Delta' \mapsto \phi^{-1}(\Delta')$ is inclusion preserving is clear. $\square$

Lemma 5.3. *Suppose Γ is a totally ordered abelian group. Then the natural order embedding $\Gamma \hookrightarrow \mathbb{Q} \otimes \Gamma$ of example 3.4 induces an order isomorphism $\mathrm{Cnv}(\mathbb{Q} \otimes \Gamma) \to \mathrm{Cnv}(\Gamma)$. Its inverse is given by $\Delta \mapsto \mathbb{Q} \otimes \Delta$.*

Proof. Since $\mathrm{Cnv}(\mathbb{Q} \otimes \Gamma)$ is totally ordered by lemma 4.5 and $\mathrm{Cnv}(\mathbb{Q} \otimes \Gamma) \to \mathrm{Cnv}(\Gamma)$ is inclusion preserving by lemma 5.2, it is sufficient to prove that $\mathrm{Cnv}(\mathbb{Q} \otimes \Gamma) \to \mathrm{Cnv}(\Gamma)$ is bijective (cf. lemma 1.5).

For injectivity, suppose that Δ and Δ' are convex subgroups of $\mathbb{Q} \otimes \Gamma$ such that $\Delta \cap \Gamma = \Delta' \cap \Gamma$. Since $\mathrm{Cnv}(\mathbb{Q} \otimes \Gamma)$ is totally ordered, we can assume without loss of generality $\Delta \subseteq \Delta'$. We will show that $\Delta = \Delta'$. Suppose $x \in \Delta'$. We can assume without loss of generality that $x \geqslant 0$. Then there exists some $n \geqslant 1$ such that $nx \in \Gamma$. Then $nx \in \Delta' \cap \Gamma = \Delta \cap \Gamma$. But Δ is divisible by lemma 4.8, so $x \in \Delta$ also.

For surjectivity, suppose Δ is a convex subgroup of Γ. Then $\mathbb{Q} \otimes \Delta$ is a convex subgroup of $\mathbb{Q} \otimes \Gamma$. To see this, suppose that $a \geqslant x \geqslant b$ in $\mathbb{Q} \otimes \Gamma$ where $a, b \in \mathbb{Q} \otimes \Delta$. There exists some $n \geqslant 1$ such that $na, nx, nb \in \Gamma$, and in fact we have $na, nb \in \Delta$. Translation invariance implies that $na \geqslant nx \geqslant nb$, so convexity of Δ tells us that $nx \in \Delta$. This means that $x \in \mathbb{Q} \otimes \Delta$.

Finally, note that $(\mathbb{Q} \otimes \Delta) \cap \Gamma = \Delta$. Indeed, clearly $\Delta \subseteq (\mathbb{Q} \otimes \Delta) \cap \Gamma$. For the reverse inclusion, suppose that $x \in (\mathbb{Q} \otimes \Delta) \cap \Gamma$ and assume without loss of generality that $x \geqslant 0$. Then $x \in \Gamma$ and there exists $n \geqslant 1$ such that $nx \in \Delta$. Translation invariance implies that

$$\cdots \geqslant nx \geqslant (n-1)x \geqslant \cdots \geqslant x \geqslant 0$$

so $x \in [0, nx]$. Convexity of Δ thus implies that $x \in \Delta$. $\square$

6 Quotients by convex subgroups

Lemma 6.1. *Suppose Γ is a totally ordered abelian group, Δ is a convex subgroup, and $\pi : \Gamma \to \Gamma/\Delta$ is the quotient map. Then there exists a unique translation invariant total order on Γ/Δ such that π is order preserving. Moreover, we have*

$$(\Gamma/\Delta)_{\geqslant 0} = \pi(\Gamma_{\geqslant 0}).$$

Proof. For existence, we will prove that the submonoid $\pi(\Gamma_{\geqslant 0})$ of Γ/Δ satisfies the conditions required to be in the image of the map in lemma 2.2. Condition (2) is clear: every element of Γ/Δ is of the form $\pi(x)$ for some $x \in \Gamma$, and either $x \geqslant 0$, in which case $\pi(x) \in \pi(\Gamma_{\geqslant 0})$, or else $-x \geqslant 0$, in which case $-\pi(x) = \pi(-x) \in \pi(\Gamma_{\geqslant 0})$. By unwinding condition (1), we find that we need to show the following: if $x \in \Gamma_{\geqslant 0}$ and there exists $y \in \Gamma_{\geqslant 0}$ such that $x + y \in \Delta$, then $x \in \Delta$ also. But notice that, by translation invariance, we have

$$x + y \geqslant x \geqslant 0$$

so convexity implies that $x \in \Delta$.

For uniqueness, suppose $\geqslant'$ is an order on Γ/Δ which makes π order preserving. We must have $\pi(\Gamma_{\geqslant 0}) \subseteq (\Gamma/\Delta)_{\geqslant' 0}$ by lemma 2.6. Then the existence proof above plus lemma 2.3 guarantee that we must have $\pi(\Gamma_{\geqslant 0}) = (\Gamma/\Delta)_{\geqslant' 0}$. □

Lemma 6.2. *Suppose Γ is a totally ordered abelian group, Δ is a convex subgroup, and $\pi : \Gamma \to \Gamma/\Delta$ is the quotient map. If $x \in \Gamma$ is such that $\pi(x) \geqslant 0$, then either $x \geqslant 0$ or $x \in \Delta$.*

Proof. Suppose that $x \lneq 0$. Since π is order preserving, we have $\pi(x) \leqslant 0$, so antisymmetry in Γ/Δ forces $\pi(x) = 0$. In other words, $x \in \ker(\pi) = \Delta$. □

Lemma 6.3. *Suppose Γ is a totally ordered abelian group, Δ is a convex subgroup, and $\pi : \Gamma \to \Gamma/\Delta$ is the quotient map. Then $\pi^{-1} : \mathrm{Cnv}(\Gamma/\Delta) \to \mathrm{Cnv}(\Gamma)$ is injective, with image consisting of all convex subgroups of Γ which contain Δ.*

Proof. Since π^{-1} defines a bijection between subgroups of Γ/Δ and subgroups of Γ containing Δ, restricting π^{-1} to $\mathrm{Cnv}(\Gamma/\Delta)$ still yields an injective map whose image is contained in the set of subgroups containing Δ. Moreover, applying π^{-1} to a *convex* subgroup yields another convex subgroup by lemma 5.2.

To complete the proof, we need to show that, if Δ' is a convex subgroup of Γ containing Δ, then Δ'/Δ is a convex subgroup of Γ/Δ. We use the criterion of lemma 4.2. Suppose we have $\pi(a) \geqslant \pi(x) \geqslant 0$ where $a \in \Delta' \cap \Gamma_{\geqslant 0}$ and $x \in \Gamma_{\geqslant 0}$. Then $\pi(a - x) \geqslant 0$. By lemma 6.2, there are two cases. First, it could be that $a - x \geqslant 0$. In this case we have $a \geqslant x \geqslant 0$, so convexity forces $x \in \Delta'$ and we are done. The second case is that $a - x \in \Delta$. But then $\pi(x) = \pi(a) \in \Delta'/\Delta$, so again we are done. □

7 Height of a totally ordered abelian group

Definition 7.1. The *height* (or *rank*) of Γ, denoted $\mathrm{ht}(\Gamma)$, is the cardinality of the set $\mathrm{Cnv}^\circ(\Gamma)$ of proper convex subgroups of Γ.

Lemma 7.2. *Suppose Γ is a totally ordered abelian group. Then $\mathrm{ht}(\mathbb{Q} \otimes \Gamma) = \mathrm{ht}(\Gamma)$.*

Proof. This follows immediately from lemma 5.3. □

Lemma 7.3. *Suppose Γ is a totally ordered abelian group and Δ a convex subgroup. Then*

$$\mathrm{ht}(\Gamma) = \mathrm{ht}(\Delta) + \mathrm{ht}(\Gamma/\Delta).$$

Proof. Let $\pi : \Gamma \to \Gamma/\Delta$ be the quotient map. The sum of two cardinals is the cardinality of the disjoint union, so we want to show that

$$\mathrm{Cnv}^\circ(\Delta) \sqcup \mathrm{Cnv}^\circ(\Gamma/\Delta)$$

bijects with $\mathrm{Cnv}^\circ(\Gamma)$. In fact, we will produce an order isomorphism

$$\mathrm{Cnv}^\circ(\Delta) + \mathrm{Cnv}(\Gamma/\Delta) \xrightarrow{f} \mathrm{Cnv}(\Gamma)$$

such that $f(\Gamma/\Delta) = \Gamma$, where the domain is the ordered sum of definition 1.9.

On $\mathrm{Cnv}(\Gamma/\Delta)$, we define f to be $\pi^{-1} : \mathrm{Cnv}(\Gamma/\Delta) \to \mathrm{Cnv}(\Gamma)$. This is order preserving from lemma 5.2. Also, if Δ' is a proper convex subgroup of Δ, it is also a convex subgroup of Γ (cf. lemma 4.4), and we define $f(\Delta')$ to be Δ' regarded as a convex subgroup of Γ. This defines the function f, which is clearly order preserving and we have $f(\Gamma/\Delta) = \Gamma$.

On the other hand, consider the function

$$\mathrm{Cnv}(\Gamma) \xrightarrow{g} (\mathrm{Cnv}(\Delta) \setminus \{\Delta\}) + \mathrm{Cnv}(\Gamma/\Delta)$$

defined by

$$g(\Delta') = \begin{cases} \Delta'/\Delta \in \mathrm{Cnv}(\Gamma/\Delta) & \text{if } \Delta' \supseteq \Delta \\ \Delta' \in \mathrm{Cnv}^\circ(\Delta) & \text{if } \Delta \supsetneq \Delta' \end{cases}$$

We know from lemma 6.3 that Δ'/Δ is in fact a convex subgroup of Γ/Δ when $\Delta' \supseteq \Delta$, so g is well defined. It is clear that g is order preserving that it is an inverse for f. □

Lemma 7.4. *Suppose Γ is a totally ordered abelian group. Then the following are equivalent.* [Bou72, VI.4, proposition 8]

(a) $\mathrm{ht}(\Gamma) \leqslant 1$. [Fuc63, IV, theorem 1]

(b) For all $a \gneq 0$ and $x \geqslant 0$ in Γ, there exists some $n \geqslant 1$ such that $na \geqslant x$.

(c) There exists an order embedding $\Gamma \hookrightarrow \mathbb{R}$.

Proof. (a) $\implies$ (b). Fix $a \gneq 0$ and $x \geqslant 0$ in Γ. Let Δ_a be the convex subgroup of Γ generated by a. By lemma 4.10, we know that

$$\Delta_a = \bigcup_{n \geqslant 1} [-na, na].$$

Then Δ_a is a nonzero convex subgroup of Γ, so, since $\mathrm{ht}(\Gamma) \leqslant 1$, we must have $\Delta_a = \Gamma$. Thus $x \in \Delta_a$, which means precisely that there exists $n \geqslant 1$ such that $na \geqslant x$.

(b) $\implies$ (c). If $\Gamma = 0$, there is nothing to show, so let us assume that $\Gamma \neq 0$. Fix an element $a \gneq 0$ in Γ. For any element $x \in \Gamma$, we define the following.

$$U(x) = \{m/n \in \mathbb{Q} \ : \ m, n \in \mathbb{Z}, n \geqslant 1, ma \geqslant nx\}$$
$$L(x) = \{m/n \in \mathbb{Q} \ : \ m, n \in \mathbb{Z}, n \geqslant 1, ma \lneq nx\}$$

The pair $(L(x), U(x))$ is a Dedekind cut in $\mathbb{Q}$, and we set

$$\phi(x) = \inf U(x) = \sup L(x).$$

Then ϕ defines an injective and order preserving homomorphism $\Gamma \to \mathbb{R}$ with the property that $\phi(a) = 1$. Details omitted. By lemma 1.5, ϕ must be an order embedding.

(c) $\implies$ (a). Observe that $\mathrm{ht}(\mathbb{R}) = 1$. Indeed, suppose Δ is a nonzero convex subgroup of $\mathbb{R}$ and $a \gneq 0$ in Δ. Let Δ_a be the convex sugbroup of Γ generated by a. Then, by lemma 4.10, we have

$$\Delta_a = \bigcup_{n \geqslant 1} [-na, na] \subseteq \Delta.$$

It is clear that every $x \in \mathbb{R}$ is contained in $[-na, na]$ for some $n \geqslant 1$, so $\mathbb{R} = \Delta_a$, forcing $\Delta = \mathbb{R}$. It follows that if there exists an order embedding $\Gamma \hookrightarrow \mathbb{R}$, then $\mathrm{ht}(\Gamma) \leqslant \mathrm{ht}(\mathbb{R}) = 1$. $\square$

8 Categorical properties

Definition 8.1. The *category of totally ordered abelian groups* is the category whose objects are totally ordered abelian groups and whose morphisms are order preserving homomorphisms.

Remark 8.2. The set of order preserving homomorphisms between two totally ordered abelian groups is a submonoid of the group of all (not necessarily order preserving) homomorphisms between them: it contains the zero map and is stable under addition, but clearly the negative of an order preserving map is *not* order preserving.

Lemma 8.3. *The category of totally ordered abelian groups has a zero object, kernels, and cokernels.*

Proof. Clearly the trivial group is a zero object. Suppose $\phi : \Gamma \to \Gamma'$ is an order preserving homomorphism. Then the group-theoretic kernel $\ker(\phi)$ with the induced order (as in example 3.3) is the kernel of ϕ in the category of totally ordered abelian groups.

Let Δ denote the convex subgroup of Γ' generated by $\phi(\Gamma)$ (i.e., the intersection of all convex subgroups containing $\phi(\Gamma)$, which is again a convex subgroup by lemma 4.6). Then the quotient map $\pi : \Gamma' \to \Gamma'/\Delta$ is a cokernel for ϕ in the category of totally ordered abelian groups. Indeed, suppose $\phi' : \Gamma' \to \Gamma''$ is an order preserving homomorphism such that $\phi' \circ \phi = 0$. Then $\phi(\Gamma) \subseteq \ker(\phi')$, but $\ker(\phi)$ is a convex subgroup by lemma 4.3, so $\Delta \subseteq \ker(\phi)$. Thus there exists a unique homomorphism $\bar{\phi}' : \Gamma'/\Delta \to \Gamma''$ such that the following diagram commutes.

$$\begin{tikzcd}
\Gamma \arrow[r, "\phi"] & \Gamma' \arrow[r, "\pi"] \arrow[dr, "\phi'"'] & \Gamma'/\Delta \arrow[d, dotted, "\bar{\phi}'"] \\
& & \Gamma''
\end{tikzcd}$$

In fact, $\bar{\phi}'$ is order preserving. We use the criterion of lemma 2.6. Suppose we have some $\pi(x) \in (\Gamma'/\Delta)_{\geqslant 0}$. Since $(\Gamma'/\Delta)_{\geqslant 0} = \pi(\Gamma_{\geqslant 0})$ by lemma 6.1, we can assume that $x \geqslant 0$. Then $\bar{\phi}'(\pi(x)) = \phi'(x) \geqslant 0$ since ϕ' is order preserving, completing the proof. □

Remark 8.4. It follows from lemma 8.3 that the category of totally ordered abelian groups has images and coimages, but it is clear from the proof of lemma 8.3 that the natural map from the coimage to the image will *not* be an isomorphism in general.

Lemma 8.5. *Suppose $\phi : \Gamma \to \Gamma'$ is an order preserving homomorphism of totally ordered abelian groups. Then the following are equivalent.*

(a) *ϕ is a monomorphism in the category of totally ordered abelian groups.*

(b) *ϕ is injective.*

Proof. Injective maps are always monomorphisms in concrete categories, so we need only to prove that monomorphisms must be injective. But notice that ϕ coequalizes the inclusion $\ker(\phi) \hookrightarrow \Gamma$ and the zero map $\ker(\phi) \to \Gamma$, so if ϕ is a monomorphism, the inclusion must be the zero map, forcing $\ker(\phi) = 0$. □

Lemma 8.6. *Suppose $\phi : \Gamma \to \Gamma'$ is an order preserving homomorphism of totally ordered abelian groups. Then the following are equivalent.*

(a) *ϕ is a normal monomorphism in the category of totally ordered abelian groups.*

(b) *ϕ is injective and $\phi(\Gamma)$ is a convex subgroup of Γ'.*

Proof. If ϕ is injective and $\phi(\Gamma)$ is a convex subgroup of Γ', then ϕ is a kernel of the projection $\pi : \Gamma' \to \Gamma'/\phi(\Gamma)$ and is therefore a regular monomorphism. Conversely, suppose ϕ is a normal monomorphism. Then ϕ is injective by lemma 8.5, and its image must be convex by lemma 4.3. □

Lemma 8.7. *Suppose $\phi : \Gamma \to \Gamma'$ is an order preserving homomorphism of totally ordered abelian groups. Then the following are equivalent.*

(a) ϕ is an epimorphism in the category of totally ordered abelian groups.

(b) The convex subgroup of Γ' generated by $\phi(\Gamma)$ is Γ'.

Proof. Let Δ denote the convex subgroup of Γ' generated by $\phi(\Gamma)$ (cf. lemma 4.6). Then ϕ equalizes the quotient map $\pi : \Gamma' \to \Gamma'/\Delta$ and the zero map $\Gamma' \to \Gamma'/\Delta$. If ϕ is an epimorphism, this means that the quotient map π is the zero map, forcing $\Gamma'/\Delta = 0$, i.e., $\Delta = \Gamma'$. Conversely, suppose $\Delta = \Gamma'$ and that ϕ equalizes two order preserving homomorphisms of totally ordered abelian groups $\alpha, \beta : \Gamma' \to \Gamma''$.

$$\begin{tikzcd} \Gamma \arrow[r, "\phi"] & \Gamma' \arrow[r, bend left, "\alpha"] \arrow[r, bend right, "\beta"'] & \Gamma'' \end{tikzcd}$$

□

Lemma 8.8. *Suppose $\phi : \Gamma \to \Gamma'$ is an order preserving homomorphism of totally ordered abelian groups. Then the following are equivalent.*

(a) ϕ is a normal epimorphism in the category of totally ordered abelian groups.

(b) ϕ is surjective.

Proof. If ϕ is surjective, then it is the cokernel of the inclusion $\ker(\phi) \hookrightarrow \Gamma$ and is therefore a normal epimorphism. Conversely, if ϕ is a normal epimorphism, it is a cokernel in the category of totally ordered abelian groups of some order preserving homomorphism α with codomain Γ. But this cokernel is precisely the quotient of Γ by the convex subgroup generated by the set-theoretic image of α, which is evidently surjective. □

Lemma 8.9. *Suppose I is a index set of cardinality at least 2, and that $(\Gamma_i)_{i \in I}$ is a family of totally ordered abelian groups indexed by I. Then neither the product nor the coproduct of this family exists in the category of totally ordered abelian groups.*

Proof. Since I has cardinality at least 2, it admits a nontrivial automorphism $\sigma : I \to I$. □

9 Hahn embedding theorem

Proposition 9.1 (Hahn embedding theorem)**.** *Let Γ be a totally ordered abelian group. Then there exists an order embedding*

$$\Gamma \hookrightarrow \prod_{\mathrm{Cnv}(\Gamma)}' \mathbb{R}.$$

References

[Bou72] Nicolas Bourbaki. *Elements of mathematics. Commutative algebra*. Translated from the French. Hermann, Paris; Addison-Wesley Publishing Co., Reading, Mass., 1972, pp. xxiv+625.

[Fuc63] László Fuchs. *Partially ordered algebraic systems*. Pergamon Press, Oxford-London-New York-Paris; Addison-Wesley Publishing Co., Inc., Reading, Mass.-Palo Alto, Calif.-London, 1963, pp. ix+229.